

Ronit Shahu

(343)-777-9638 | Ottawa, Ontario (open to relocate) | ronitshahu@gmail.com | [LinkedIn](#) | [Website](#) | [GitHub](#) | [Live Project](#) |

MACHINE LEARNING ENGINEER

PROFESSIONAL SUMMARY

Focused on building compliant, interpretable ML solutions for regulated sectors like banking, healthcare, and public research. Data-driven Machine Learning Engineer specializing in building and deploying production-grade ML systems for finance and healthcare. Proficient in Python, scikit-learn, PyTorch, FastAPI, Docker, and MLflow with hands-on experience deploying APIs on AWS and familiarity with Azure ML and Databricks (PySpark). Skilled in developing feature pipelines, model-serving microservices, and explainable ML workflows aligned with governance standards. Delivered measurable business impact reduced manual operations by 40%, cut infrastructure costs by 35%, and accelerated analytics cycles by integrating CI/CD + monitoring into scalable ML pipelines.

AREA OF EXPERTISE

Python	PyTorch & Tensorflow	Feature Engineering	Model Evaluation & Explainability
SQL	NLP & Generative AI (LLMs)	PySpark	AWS (EC2, Lambda)
GIT / CI/CD	A/B Training / Testing	Pandas & Numpy	Databricks
Kubernetes	RESTAPI – FastAPI / Flask	Azure ML	Scikit - Learn

CAREER HIGHLIGHTS

AI-Driven Process Transformation: Delivered scalable, production-grade AI systems that reduced manual review time by 40%, improved data quality by 35%, and accelerated analytics pipelines across healthcare and financial workflows through automated data validation and model-driven insights.

End-to-End ML Systems Delivery & Leadership: Progressed from developer to technical lead, managing a 15+ cross-functional engineering team to deploy containerized ML APIs (FastAPI + Docker) with MLflow and CI/CD integration, enhancing model accuracy by 30% and enabling repeatable, compliant MLOps practices.

Continuous Improvement & Collaboration: Recognized as a Top Talent and Global AI Hackathon finalist for pioneering applications of LLM-based NLP and automated document intelligence, demonstrating a commitment to innovation, explainability, and cross-functional collaboration in applied AI delivery.

WORK EXPERIENCE

Lead Machine Learning Researcher, Canadian Robotics and Artificial Intelligence Ethical Design Lab, University of Ottawa, ON | Jan 2025 – May 2025

Appointed to lead a cross-functional team contributing to national AI ethics research.

- Led a 15-member data science team delivering ML-driven automation systems for national AI-ethics research, overseeing the full lifecycle from data engineering to model deployment.
- Built and deployed containerized ML APIs using FastAPI + Docker, hosted on AWS EC2 and reproducible on Azure ML Workspaces with integrated MLflow experiment tracking and CI/CD pipelines for versioned model delivery.
- Designed feature pipelines and preprocessing workflows using Pandas / PySpark (Databricks-compatible) for real-time inference and analytics over structured & unstructured data.
- Deployed scalable microservices with automated scheduling and monitoring, improving cost efficiency by 50% and ensuring compliance with performance KPIs. Contributed to AI ethics research by developing transparent and auditable model reporting mechanisms aligned with explainable AI standards.

Software Developer – Life science and Health Care, Persistent Systems Limited, India | Dec 2021 – July 2023

- AI-powered Certification Verification System: Automated image & PDF validation pipeline using machine learning and OCR deployed via Azure Machine Learning pipelines, reducing manual verification by 70% and operational costs by 35%.

- Medical Data Extraction for Clinical Trials: Designed and trained custom spaCy NER models to extract cancer-related entities, integrated with Azure ML endpoints for deployment, reducing data preparation time by 40% and improving insight accuracy.
- Healthcare NLP Microservices: Developed and maintained Flask / FastAPI-based APIs containerized with Docker and orchestrated through Azure ML + CI/CD, improving information retrieval speed by 12% and enabling consistent model retraining.
- Feature Engineering & Data Pipelines: Engineered features from unstructured clinical text and structured EHR data (20K+ records) using Pandas / PySpark, optimizing preprocessing for downstream ML workflows.
- Cross-functional Collaboration: Partnered with data scientists, DevOps, and clinical SMEs across global teams to translate healthcare data challenges into deployable, compliant ML solutions aligned with responsible AI practices

EDUCATION

Master of System Science and Engineering (Interdisciplinary Artificial Intelligence) University of Ottawa, ON	2023 – 2025
Bachelor of Engineering in Electronics Engineering Shri Ramdeobaba College of Engineering and Management, India	2018 – 2022

AWARDS AND ACHIEVEMENTS

• Research Fellowship Scholarship, University of Ottawa, ON	2025
• Elected as Program Co-Ordinator for Systems Science & Eng. as a liaison between students and faculty.	2023
• Top Talent 2023, Persistent Systems (for measurable ML project impact), Persistent Systems Limited, India	2023
• 2nd Runner-Up, Global AI Hackathon Semicolons 2023, India	2023

CERTIFICATIONS

• In Progress: AWS Certified AI Practitioner	(Expected Q1 2026)
• Microsoft Azure AZ-900 Fundamentals	2022

PUBLICATIONS

• <u>Audio + Text based Emotion Recognition System</u>	2021
• <u>Detection of COVID-19 using Machine Learning and Android App Development</u>	2021

RELEVANT PROJECTS

Cancer-Aware Medical Chatbot with Research Recommender (<i>University of Ottawa, 2025</i>)	
• Designed and deployed an LLM-orchestrated GenAI chatbot using Dialogflow ES, FastAPI, and Google Gemini LLM, integrating TF-IDF, Doc2Vec, and K-Means similarity modules to recommend cancer research papers from Mayo Clinic datasets. Optimized response latency by 45% through prompt engineering, multi-intent routing, and API parallelization—delivering faster, context-aware medical information retrieval.	
• Implemented modular LLM orchestration combining Gemini outputs with structured data from UMLS and Mayo Clinic APIs, demonstrating explainable retrieval-augmented generation and hybrid search workflows.	
Fraud Detection & Risk Prioritization (PaySim)	
• Built a real-time fraud-detection pipeline using scikit-learn, advanced feature engineering, and ensemble classifiers on synthetic transaction data. Deployed as REST APIs via FastAPI + AWS Lambda for low-latency inference, enabling scalable model-serving and automated risk scoring for streaming data	
Credit-Risk Modeling & Explainable AI Pipeline (German Credit Dataset)	
• Developed an end-to-end, cost-sensitive ML pipeline replicating enterprise credit-risk workflows using Python (scikit-learn, SHAP, MLflow). Engineered interpretable features with Weight-of-Evidence (WOE) / Information Value (IV) analysis, applied fairness audits, and encoded ordinal/nominal variables for Basel-aligned scorecards.	
• Benchmarked Logistic Regression, Random Forest, and Gradient Boosting models (AUC ≈ 0.83) while minimizing expected misclassification cost (Bad → Good penalty 5×). Integrated SHAP explainability for global & local feature insights—ensuring transparency, bias detection, and audit-ready governance.	